

Saket Deshmukh

AI/GenAI Engineer — Portfolio

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SUMMARY

AI/GenAI Engineer with **2.5+ years of experience** building and deploying **production LLM applications**, including RAG systems, agentic workflows, and AI APIs. Experienced with **Python, LangChain/LangGraph, vector databases, LLM evaluation, and cloud deployment**.

SKILLS

Programming: Python, SQL, JavaScript, Bash

AI / ML: LLMs, Generative AI, Fine-Tuning, Embeddings, Structured Output

LLM Engineering: RAG, Agentic AI, Prompt Engineering, Grounding, Guardrails, LLM Evaluation

Agent Frameworks: LangChain, LangGraph, Tool Calling, MCP, A2A

API Development: FastAPI, REST APIs, OpenAI API, Pydantic

Retrieval: FAISS, pgvector, BM25, Vector Search, Hybrid Search, Semantic Search, RRF

Evaluation & Observability: LLMops, LangSmith, TruLens, LLM Evaluation, Tracing, Monitoring

Cloud & Deployment: AWS, Azure, Docker, Kubernetes, Terraform, GitHub Actions

Databases: PostgreSQL, MySQL

EXPERIENCE

KWAD Corporation Pvt. Ltd. — AI Engineer

Apr 2025 – Present — Pune, India

Grounded AI Platform for Clinical Record Processing

- Built a **full-stack AI platform** in **Python (FastAPI, Pydantic v2, async SQLAlchemy, PostgreSQL)** that extracts and reasons over clinical documents – scans, spreadsheets, handwriting, multi-language records for embryologists and lab managers.
- Designed a **multi-provider LLM orchestration layer** via **LiteLLM** with schema-forced structured output, **RAG** using **pgvector**, and vision-model transcription for scanned/handwritten documents.
- Implemented a **deterministic grounding-verification layer** validating every model output against its source span; deployed on **AWS ECS** with **Docker**, **S3**, and **CI** (mypy, ruff, pytest) coverage gates.

AI Compliance & Financial Operations Copilot

- Built an **agentic RAG system (LangChain, LangGraph)** for natural-language investigation of B2B transaction and compliance data across **12 tenants**, with conditional routing, tool calling, and human-in-the-loop escalation.
- Built the **RAG pipeline** end to end (chunking, embeddings, **pgvector**) with **grounded, cited responses**, reducing hallucinated answers by **38%**; **fine-tuned** models via OpenAI's API for **99% schema compliance**.
- Deployed via **Docker/Kubernetes**; added tracing and **evaluation/monitoring** metrics supporting **9 iterations** of prompt/retrieval improvements, cutting mean-time-to-detect regressions by **45%**.

Briowin Private Limited — Software Engineer

Feb 2024 – Mar 2025 — Mumbai, India

CatalogSage – AI-Powered Product Catalog Assistant

- Built an **AI-powered RAG assistant** for natural-language search and recommendations across a **300+ SKU catalog (Python, FastAPI, MCP, OpenAI API)**, cutting query resolution time by **52%**.
- Implemented a **LangGraph agent workflow** with conditional routing for input validation, hybrid retrieval, and graceful fallback, improving end-to-end request success rate to **94%**.
- Tuned **hybrid retrieval (OpenAI embeddings, FAISS, BM25 via RRF)**, improving top-k retrieval accuracy by **27%**; built a **TruLens** evaluation pipeline catching **18%** of regressions pre-release.

PROJECTS

AI-Powered SQL Query Optimizer

Python, FastAPI, LangGraph, OpenAI API, SQLGlot, PostgreSQL, Docker

- Built an AI-driven SQL optimization pipeline (**LangGraph, OpenAI API, SQLGlot**) analyzing PostgreSQL query plans to generate indexing/rewrite recommendations, with a **safety-validation layer** verifying correctness on a **500K+ row** dataset.

Self-Hosted LLM Inference Platform

AWS EKS, Kubernetes, Terraform, Docker, vLLM

- Deployed a **self-hosted LLM inference platform** on **AWS EKS** using **Docker** and **vLLM** for model serving, with **Terraform** provisioning VPC, IAM, and Kubernetes infrastructure.

EDUCATION

B.E. in Computer Engineering

2019 – 2023 — Pune, India

Savitribai Phule Pune University